

100 Days of Azure DevOps

Day 82 Handout — One contract beats fourteen slightly different broken builds

9 - Advanced & Enterprise · Personal lab only · Educational content · Not a sales pitch

Architecture A — reuse knobs in Azure Pipelines YAML

Knob	YAML	Use today
Step template	- template: templates/build.yml	Extract restore/build/test once
parameters	parameters: - name: projectPath type: string	The contract. No secret defaults
extends	extends: template: templates/pipeline.yml	Literacy: template owns stages; lab can stay steps
Template repo ref	resource: repo + ref: refs/tags/v1	Later; same repo templates/ is enough
Each: / insert	\${{ each p in parameters.projects }}	Skip until two consumers exist (Day 90)

Architecture B — extract vs leave inline

Smell	Extract	Leave inline
Same npm ci / dotnet publish in 2 files	templates/build.yml	One-off pipeline you will delete tonight
Node 20 vs 22 pin	One parameter nodeVersion	Do not copy the pin into 14 files
Env-specific deploy	Keep deploy in the root pipeline or a deploy.yml	Do not hide env names as secret defaults
extends vs steps	steps template first	extends when you need a locked stage graph
Clever defaults	Required parameters	A default password or PAT in the template

Architecture C — pitfalls that fake reuse

Pitfall	What you see	Fix
Copy-paste 'standard'	14 pipelines, 14 forgotten cache keys	One template, many callers
Rearranged, not extracted	Still duplicated restore/build after 'cleanup'	Call site is one template line + path
Secret in parameters default	PAT sitting in templates/	Variable group / Key Vault; required param
Unversioned remote template	@main of a moving repo	Same-repo path or a tag ref
Abstract novel on day one	20 layers, zero green runs	echo \${{ parameters.projectPath }} is a legal first cut

One-liner to remember

Copy-paste YAML is how organizations invent 14 slightly different ways to be broken.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Azure DevOps project azure-100-labs. Templates live in the same repo. Microsoft-hosted ubuntu-latest.

- 1 Create templates/build.yml with a required parameter projectPath (string) and a step that builds or echoes that path.
- 2 Point the main pipeline at it: steps: - template: templates/build.yml with parameters.projectPath.
- 3 Run the pipeline once on ubuntu-latest. The log must show the parameter value.
- 4 If two sample apps already exist, call the same template twice with different paths. If not, one caller today; two consumers on Day 90.

- 5 Search the repo for duplicated npm ci / dotnet publish / docker build. If still duplicated, you rearranged — extract the rest.
- 6 Write docs/templates-day82.md: contract (parameter names) and the rule 'Node version changes in one file'.

templates/build.yml plus a boring call site

```
# templates/build.yml
parameters:
- name: projectPath
  type: string
steps:
- script: echo Building ${ parameters.projectPath }
  displayName: Build ${ parameters.projectPath }

# azure-pipelines.yml (call site)
pool:
  vmImage: ubuntu-latest
steps:
- template: templates/build.yml
  parameters:
    projectPath: src/demo-shipboard
```

Done checklist

- templates/build.yml exists with a required projectPath
- Main pipeline calls the template; log shows the path
- No secret defaults in the template
- Call site is boring (one template line)
- Posted the LinkedIn document

Tomorrow — Day 83

Marketplace extensions — a pinch of spice, or a written reason to install zero